NOETHER–LEFSCHETZ VANISHING FOR HYPERSURFACES IN $\mathbb{P}^5$ AS AN EXPLICIT HYPOTHESIS: A MACHINE-CHECKED LEAN 4 INTERFACE

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Abstract. We record in Lean 4, with no user axioms, the propositional form of Noether–Lefschetz vanishing: a very general smooth hypersurface of degree $d \geq 3$ in $\mathbb{P}^5$ has no rational Hodge class in H^4 beyond multiples of h^2. Hypersurfaces are abstract records with a degree, a "very general" predicate and an "extra class" predicate. Conditional on the vanishing sentence, we prove that an extra class on such a hypersurface of degree at least 3 gives a contradiction, and that there is no extra class when the degree is at least 6. A second encoding over an abstract moduli setting proves that the $d \geq 3$ sentence implies its $d \geq 6$ case. The predicate $\mathrm{Hdg}^2 = \mathbb{Q}h^2$ in that encoding is a placeholder defined as `True`. All results are conditional. Beauville's monodromy theorem and Deligne's invariance argument are not formalized.

1. Introduction

For a very general smooth hypersurface $X \subset \mathbb{P}^5$ of degree $d \geq 3$, the space of rational Hodge classes $H^4(X, \mathbb{Q}) \cap H^{2,2}(X)$ is spanned by h^2. The standard proof combines Beauville's theorem that monodromy is Zariski dense in the orthogonal group of vanishing cohomology [1] with Deligne's theorem that a Hodge class on a very general fibre is monodromy invariant [2]. For $d = 2$ the statement fails, because smooth quadric fourfolds contain planes. Neither argument is formalized here. The repository previously asserted the vanishing (for $d \geq 6$) and the moduli-level statement (for $d \geq 3$) as Lean axioms, and one file did not compile. In this version every axiom is removed. The vanishing is a hypothesis, and all results are implications from it.

2. Encoding by records (`NoetherLefschetz.Basic`)

Definition 2.1 (`Hypersurface`, `ExtraClass`, `VanishingOfExtraClasses`)**.** A *hypersurface record* X has a degree $\deg X \in \mathbb{N}$ and propositions `isVeryGeneral` and `hasExtraClass`. `ExtraClass X` is `hasExtraClass`. The *vanishing sentence* for X is: if $3 \leq \deg X$ and X is very general, then X has no extra class. The proposition `noether_lefschetz X` is defined to be this sentence. `HodgeLocusPosAlgebraic d` is defined to be $6 \leq d$.

Theorem 2.2 (Conditional; `no_extra_on_nl_locus`)**.** *Let X satisfy the vanishing sentence. If $3 \leq \deg X$, X is very general and X has an extra class, then* `False`*.*

Theorem 2.3 (Conditional; `noether_lefschetz_of_vanishing`)**.** *Let X satisfy the vanishing sentence. If $6 \leq \deg X$ and X is very general, then X has no extra class.*

3. Encoding over a moduli setting (`NoetherLefschetz.Spec`)

Definition 3.1 (`SpecSetting`, `hdg2_eq_Qh2`, `extra_rational_classes_vanish`)**.** A *spec setting* S consists of types M_d for $d \in \mathbb{N}$ (moduli of smooth degree-d hypersurfaces) and a predicate `IsVeryGeneral` on properties of points of M_d. The predicate `hdg2_eq_Qh2` is a placeholder *defined as* `True`. The sentence `extra_rational_classes_vanish d` is: $3 \leq d$ implies `IsVeryGeneral`$(X \mapsto$ `hdg2_eq_Qh2` $d\, X)$.

Proposition 3.2 (`extra_rational_classes_vanish_unfolds`)**.** *For every S and d,* `extra_rational_classes_vanish d` *equals its defining formula.*

Date: November 4, 2026.

Theorem 3.3 (Conditional; `d_ge_6_subcase`)**.** *Assume that for all $d \geq 3$,* `IsVeryGeneral`$(X \mapsto$ `hdg2_eq_Qh2` $d\,X)$*. Then the same holds for every $d \geq 6$.*

Remark 3.4. Because `hdg2_eq_Qh2` is `True`, the second encoding carries no Hodge-theoretic content. It records only how the threshold $d \geq 6$ follows from $d \geq 3$.

4. Correspondence between claims and Lean declarations

Every mathematical claim of this paper is listed below with the Lean declaration that proves it. "Conditional" means the hypothesis named in the statement is an explicit argument of the Lean theorem and is *not* proved in the repository. No claim in this paper goes beyond these statements.

Claim	Lean declaration	Status
Vanishing, deg ≥ 3, very general, extra class $\Rightarrow$ False	`NoetherLefschetz.no_extra_on_nl_locus`	Conditional on `VanishingOfExtraClasses X`
Vanishing $\Rightarrow$ no extra class for deg ≥ 6	`NoetherLefschetz.noether_lefschetz_of_vanishing`	Conditional on `VanishingOfExtraClasses X`
Sentence unfolds to its definition	`NoetherLefschetz.Spec.extra_rational_classes_vanish_unfolds`	Proved (by definition)
$d \geq 3$ sentence $\Rightarrow$ $d \geq 6$ case	`NoetherLefschetz.Spec.d_ge_6_subcase`	Conditional on the $d \geq 3$ hypothesis

5. Scope and limitations

Hodge theory, monodromy, moduli of hypersurfaces and the theorems of Beauville and Deligne are not formalized. Hypersurface records and spec settings are arbitrary data, and `hdg2_eq_Qh2` is `True`. The Noether–Lefschetz theorem is not proved; the results are implications from it. The bound 6 is not claimed to be sharp. Statement changes in this version: the axioms `ExtraClass` and `noether_lefschetz` in `Basic` are removed. `Basic` did not compile and has been repaired: `hasExtraClass` is a record field, and `ExtraClass` and `noether_lefschetz` are definitions. The old axiom `noether_lefschetz` ($d \geq 6$) is now the conditional theorem `noether_lefschetz_of_vanishing`. In `Spec`, the axioms `SmoothHypersurfaceModuli` and `IsVeryGeneral` are now fields of `SpecSetting`, and the axiom `extra_rational_classes_vanish_holds` is now a hypothesis of `d_ge_6_subcase`.

Appendix A. Lean source

The complete Lean source of the certified development (copyright headers omitted) is reproduced verbatim.

NoetherLefschetz/Basic.lean.

```
/-!
# Noether–Lefschetz as named propositions

Not the Hodge conjecture. Literature, not a proof in this file.

Geometric range: a very general smooth hypersurface of degree `d ≥ 3` in `ℙ⁵`
has no extra rational Hodge class. Degree `2` fails (quadric fourfolds contain planes).
No `axiom`. No `sorry`. No dummy `P ∧ ¬P`.
`VanishingOfExtraClasses` is a sentence. A term of that sentence would be
Beauville (monodromy dense in the orthogonal group of vanishing cohomology)
plus Deligne (a rational Hodge class on a very general fibre is
monodromy-invariant). Those arguments are not in this file; every result is
conditional on that sentence, supplied as a hypothesis.
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```
-/

namespace NoetherLefschetz

/-- A named hypersurface in ℙ⁵, as a record.
`hasExtraClass` is the geometric predicate “Hdg² larger than ℚ h²”.
It is a field, not `degree = 0 ∧ degree ≠ 0`. -/
structure Hypersurface where
  degree : Nat
  isVeryGeneral : Prop
  hasExtraClass : Prop

/-- Extra means not a rational multiple of h². -/
def ExtraClass (X : Hypersurface) : Prop :=
  X.hasExtraClass

/-- Vanishing sentence (monodromy + Deligne).
Very general smooth hypersurface in ℙ⁵ of degree ≥ 3. -/
def VanishingOfExtraClasses (X : Hypersurface) : Prop :=
  3 ≤ X.degree → X.isVeryGeneral → ¬ ExtraClass X

/-- BKU sentence, as a name only: the inequality `6 ≤ d`. -/
def HodgeLocusPosAlgebraic (d : Nat) : Prop :=
  6 ≤ d

/-- Former axiom name, now a proposition. -/
def noether_lefschetz (X : Hypersurface) : Prop :=
  VanishingOfExtraClasses X

/-- Implication from the named vanishing sentence. Does not inhabit that sentence. -/
theorem no_extra_on_nl_locus (X : Hypersurface)
    (hV : VanishingOfExtraClasses X)
    (hd : 3 ≤ X.degree) (hvg : X.isVeryGeneral)
    (hExtra : ExtraClass X) : False :=
  hV hd hvg hExtra

/-- The former axiom `noether_lefschetz` (degree ≥ 6), now conditional on the
vanishing sentence. -/
theorem noether_lefschetz_of_vanishing (X : Hypersurface)
    (hV : VanishingOfExtraClasses X)
    (hd : 6 ≤ X.degree) (hvg : X.isVeryGeneral) :
    ¬ ExtraClass X :=
  hV (Nat.le_trans (by decide) hd) hvg

end NoetherLefschetz
```

NoetherLefschetz/Spec.lean.

```
/-
Statement encoding. No Mathlib imports.
The former axioms `SmoothHypersurfaceModuli`, `IsVeryGeneral` are fields of
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```
`SpecSetting`; the former axiom `extra_rational_classes_vanish_holds` is a hypothesis.
Does not prove Beauville, Deligne, or Noether-Lefschetz.
Note: `hdg2_eq_Qh2` is a placeholder defined as `True`.
-/

namespace NoetherLefschetz
namespace Spec

/-- Abstract moduli of smooth degree-`d` hypersurfaces in P^5 and a
"very general" predicate. Supplied by the user. -/
structure SpecSetting where
  SmoothHypersurfaceModuli : Nat → Type
  IsVeryGeneral : {d : Nat} → (SmoothHypersurfaceModuli d → Prop) → Prop

variable (S : SpecSetting)

/-- H^4(X,Q) ∩ H^{2,2}(X) = Q h^2. Placeholder: defined as `True`. -/
def hdg2_eq_Qh2 (d : Nat) (_X : S.SmoothHypersurfaceModuli d) : Prop :=
  True

/-- The sentence. Proof is not supplied. -/
def extra_rational_classes_vanish (d : Nat) : Prop :=
  3 ≤ d → S.IsVeryGeneral fun X => hdg2_eq_Qh2 S d X

theorem extra_rational_classes_vanish_unfolds (d : Nat) :
    extra_rational_classes_vanish S d =
      (3 ≤ d → S.IsVeryGeneral fun X => hdg2_eq_Qh2 S d X) :=
  rfl

/-- d ≥ 6 is a subcase of the d ≥ 3 sentence (conditional on that sentence). -/
theorem d_ge_6_subcase
    (extra_rational_classes_vanish_holds : ∀ d, 3 ≤ d → S.IsVeryGeneral fun X => hdg2_eq_Qh2 S d X)
    (d : Nat) (hd : 6 ≤ d) :
    S.IsVeryGeneral fun X => hdg2_eq_Qh2 S d X :=
  extra_rational_classes_vanish_holds d (Nat.le_trans (by decide) hd)

end Spec
end NoetherLefschetz
```

Appendix B. Verification certificate

The formal development was machine-checked as follows.

- Repository:
  `https://github.com/BenFrohman/NoetherLefschetz`
- Verified commit:
  `ec7289969d31f50bf84395a515875466a656422d`
- Lean toolchain: `leanprover/lean4:v4.34.0`
- Mathlib: none (the development depends only on Lean core)
- Continuous integration run (build, axiom audit, `sorry`/`axiom` scan; passed):
  `https://github.com/BenFrohman/NoetherLefschetz/actions/runs/41794912612`

- SHA-256 of the GitHub source tarball of the verified commit, downloaded twice with identical result:

  `https://github.com/BenFrohman/NoetherLefschetz/archive/ec7289969d31f50bf84395a515875466a656422d.tar.gz`

  `43d91047d05a339cb5ce78982e0f84b4d44582dadeea0a82efccd5db78dc565f`

The tarball hash refers to the verified commit rather than to a release tag, since a document cannot contain the hash of an archive that includes itself. The Lean sources are unchanged between that commit and the release tag.

The development contains no `sorry` and no user-declared axioms. Every one of the 4 theorems and lemmas listed below depends only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound` (or on none). The table is the verbatim output of `lake env lean AxiomAudit.lean` (`#print axioms`) on the verified commit.

Lean declaration	`#print axioms`
`NoetherLefschetz.no_extra_on_nl_locus`	`(no axioms)`
`NoetherLefschetz.noether_lefschetz_of_vanishing`	`(no axioms)`
`NoetherLefschetz.Spec.d_ge_6_subcase`	`(no axioms)`
`NoetherLefschetz.Spec.extra_rational_classes_vanish_unfolds`	`(no axioms)`

References

[1] A. Beauville, *Le groupe de monodromie des familles universelles d'hypersurfaces et d'intersections complètes*, Lecture Notes in Math. 1194, Springer (1986), 8–18.

[2] P. Deligne, *Théorie de Hodge II*, Publ. Math. IHÉS 40 (1971), 5–57.

[3] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, CADE-28, LNCS 12699 (2021), 625–635.

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URL: `https://github.com/BenFrohman/NoetherLefschetz`